Expectation Maximization: From Mixture Densities to a Provably Convergent Algorithm

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Lecture notes and tutorial

Abstract

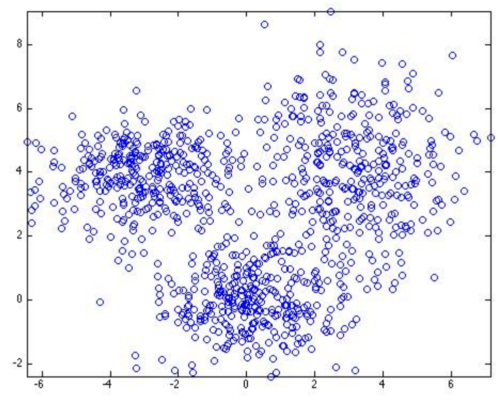
Data with several clusters cannot be described by a single Gaussian, and the natural remedy — a mixture of Gaussians — produces a log-likelihood that neither yields a closed-form maximum nor suits plain gradient ascent, since the objective is non-concave. These notes develop the Expectation-Maximization (E-M) algorithm as the standard answer. After reviewing convexity and Jensen's inequality, we state the E-M update and prove its central guarantee: each iteration leaves the log-likelihood non-decreasing. The proof is given twice — once through the ELBO–KL decomposition familiar from variational inference, viewing E-M as maximization-maximization, and once directly through the classical Q/H decomposition, where a one-line Jensen argument shows the cross-entropy term can never decrease. The framework is then applied in full to the Gaussian Mixture Model: responsibilities form the E-step, and Lagrange multipliers plus matrix calculus give closed-form M-step updates for the mixing weights, means, and covariances, each a responsibility-weighted average. A sanity check confirms the updates collapse to the single-Gaussian maximum-likelihood estimates when $k = 1$. Exercises with worked solutions conclude the notes.

Keywords: expectation maximization; Gaussian mixture model; latent variables; Jensen's inequality; ELBO; maximum likelihood

Contents

1 Motivation: Mixture Density Models

When you have data that looks like:



Can you fit them using a single-mode Gaussian distribution, i.e.,:

$$
\begin{aligned}
p(x) &= \mathcal{N}(x|\mu, \Sigma) \\
&= (2\pi)^{-k/2}|\Sigma|^{-\frac{1}{2}} \exp\Big(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big)
\end{aligned} \tag{1}
$$

Clearly not! Data like this is typically modelled using mixture densities [1] — in this case, a Gaussian Mixture Model with k mixtures (GMM):

$$
p(x) = \sum_{l=1}^{k} \alpha_l \mathcal{N}(x|\mu_l, \Sigma_l) \qquad \text{s.t.} \quad \sum_{l=1}^{k} \alpha_l = 1 \tag{2}
$$

1.1 Gaussian Mixture model result

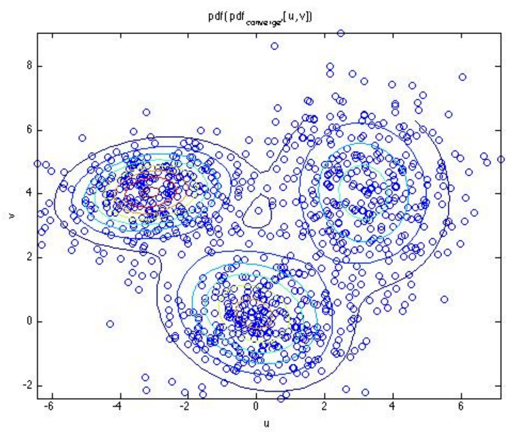


Figure 1: GMM fitting result

Let $\Theta = \{\alpha_1, \dots \alpha_k, \mu_1, \dots \mu_k, \Sigma_1, \dots \Sigma_k\}$

$$\begin{aligned}\Theta_{\text{MLE}} &= \arg\max_{\Theta} \mathcal{L}(\Theta|X) \\ &= \arg\max_{\Theta} \left(\sum_{i=1}^{n} \log \sum_{l=1}^{k} \alpha_l \mathcal{N}(x_i|\mu_l, \Sigma_l) \right) \qquad \sum_{l=1}^{k} \alpha_l = 1\end{aligned} \tag{3}$$

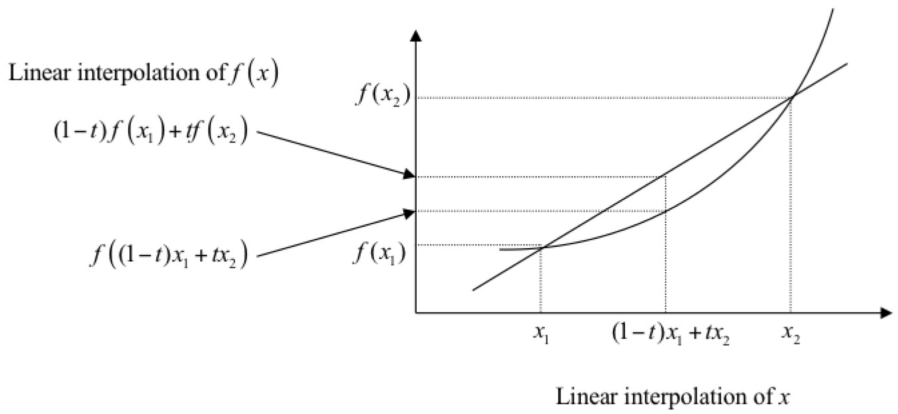
1. Unlike the single-mode Gaussian, we cannot simply take derivatives and set them to zero, i.e., optimize it analytically.
2. In terms of optimizing its parameters: the problem is **non-concave**. So traditional gradient ascent is **not** suitable for it, i.e., there are many local maxima.

The **goal** is to find an iterative process that ensures that $\log(p(X|\Theta^{(g)}))$ remains non-decreasing for $g = 1, \dots$.

To do so, for data $X = \{x_1, \dots x_n\}$, we introduce "latent" variables $Z = \{z_1, \dots z_n\}$, each z_i indicating which mixture component x_i belongs to. The core problem then simply boils down to fitting a single Gaussian.

2 Preliminaries

2.1 Convex function



$$f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2) \qquad t \in [0, 1] \tag{4}$$

2.2 Jensen's inequality

Using notation ϕ instead of f:

$$\phi((1-t)x_1 + tx_2) \leq (1-t)\phi(x_1) + t\phi(x_2) \qquad t \in [0, 1] \tag{5}$$

Can be generalized further for any convex combination, i.e., let $\sum_{i=1}^{n} p_i = 1$:

$$\begin{aligned}\phi(p_1x_1 + p_2x_2 + \dots + p_nx_n) &\leq p_1\phi(x_1) + p_2\phi(x_2) + \dots + p_n\phi(x_n) \qquad \sum_{i=1}^{n} p_i = 1 \\ \implies \phi\Big(\sum_{i=1}^{n} p_i x_i\Big) &\leq \sum_{i=1}^{n} p_i \phi(x_i) \\ \implies \phi\Big(\sum_{i=1}^{n} p_i f(x_i)\Big) &\leq \sum_{i=1}^{n} p_i \phi(f(x_i)) \qquad \text{by replacing } x_i \text{ with } f(x_i)\end{aligned} \tag{6}$$

generalize to the continuous case:

$$\phi\left(\int_x f(x)p(x)\mathrm{d}x\right) \leq \int_x \phi(f(x))p(x)\mathrm{d}x \implies \phi\big(\mathbb{E}[f(x)]\big) \leq \mathbb{E}[\phi(f(x))] \tag{7}$$

2.2.1 Jensen's inequality example: $-\log(x)$

$\phi(x) = -\log(x)$ is a convex function:

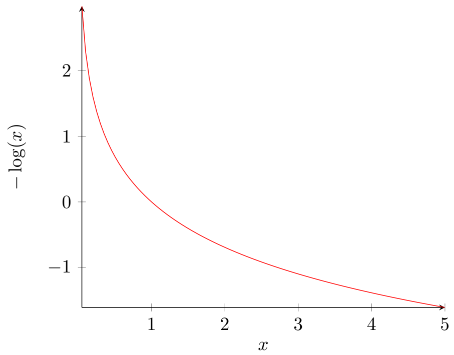


Figure 2: example of convex function $-\log(x)$

1. when $\phi(.)$ is convex:

$$\begin{aligned} &\phi(\mathbb{E}[f(\mathbf{x})]) \leq \mathbb{E}[\phi(f(\mathbf{x}))] \\ \text{e.g.} \quad &-\log(\mathbb{E}[f(\mathbf{x})]) \leq \mathbb{E}[-\log(f(\mathbf{x}))] \end{aligned} \tag{8}$$

2. when $\phi(.)$ is concave:

$$\begin{aligned} &\phi(\mathbb{E}[f(\mathbf{x})]) \geq \mathbb{E}[\phi(f(\mathbf{x}))] \\ \text{e.g.} \quad &\log(\mathbb{E}[f(\mathbf{x})]) \geq \mathbb{E}[\log(f(\mathbf{x}))] \end{aligned} \tag{9}$$

3 Expectation-Maximization Algorithm

Instead of performing:

$$\begin{aligned} \Theta^{\text{MLE}} &= \arg\max_{\Theta} \mathcal{L}(\Theta|X) \\ &= \arg\max_{\Theta} (\log[p(X|\Theta)]) \end{aligned} \tag{10}$$

The trick is to introduce some "latent" variable Z into the model.
For each iteration of the E-M algorithm [2], we perform:

$$\Theta^{(g+1)} = \arg\max_{\Theta} \left(\int_Z \log\left(p(X,Z|\Theta)\right)p(Z|X,\Theta^{(g)})\mathrm{d}Z\right) \tag{11}$$

However, before we apply it, we must ensure convergence [3], i.e.,

$$
\begin{aligned}
\log(p(X|\Theta^{(g+1)})) &= \mathcal{L}(\Theta^{(g+1)}|X) \\
&\geq \mathcal{L}(\Theta^{(g)}|X) \\
&= \log(p(X|\Theta^{(g)})) \quad \forall g
\end{aligned}
\tag{12}
$$

Note the difference between the variable Θ and the constant $\Theta^{(g)}$. Also note that gradient ascent is **not** suitable for many E-M problems, as they can be **non-concave**.

4 Proof of convergence: Maximization-Maximization

We have seen this in the variational-inference literature [4], as the ELBO-KL decomposition:

$$
\begin{aligned}
\mathcal{L}(\Theta|X) &= \log\left(p(X|\Theta)\right) \\
&= \log\left(\frac{p(X,Z|\Theta)}{p(Z|X,\Theta)}\right) \\
&= \log\left(\frac{p(X,Z|\Theta)}{q(Z)} \times \frac{q(Z)}{p(Z|X,\Theta)}\right) \\
&= \log\left(\frac{p(X,Z|\Theta)}{q(Z)}\right) + \log\left(\frac{q(Z)}{p(Z|X,\Theta)}\right) \\
&= \int_Z \log\left(\frac{p(X,Z|\Theta)}{q(Z)}\right) q(Z)\mathrm{d}Z + \int_Z \log\left(\frac{q(Z)}{p(Z|X,\Theta)}\right) q(Z)\mathrm{d}Z \\
&= \text{ELBO}(\Theta, q) + \text{KL}\big(q(Z)\|p(Z|X,\Theta)\big)
\end{aligned}
\tag{13}
$$

or to use Jensen's inequality:

$$
\begin{aligned}
\mathcal{L}(\Theta|X) = \log p(X|\Theta) &= \log \int_Z p(X,Z|\Theta)\mathrm{d}Z \\
&= \log\Big(\int_Z \frac{p(X,Z|\Theta)}{q(Z)} q(Z)\mathrm{d}Z\Big) \\
&\geq \int_Z \log\Big(\frac{p(X,Z|\Theta)}{q(Z)}\Big) q(Z)\mathrm{d}Z \\
&= \text{ELBO}(\Theta, q)
\end{aligned}
\tag{14}
$$

4.1 Maximization of ELBO

When applying E-M as a Maximization-Maximization algorithm, we only need to optimize ELBO, let's revisit the ELBO-KL decomposition again:

$$
\begin{aligned}
\mathcal{L}(\Theta|X) &= \int_Z \log\left(\frac{p(X,Z|\Theta)}{q(Z)}\right) q(Z)\mathrm{d}Z + \int_Z \log\left(\frac{q(Z)}{p(Z|X,\Theta)}\right) q(Z)\mathrm{d}Z \\
&= \text{ELBO}(\Theta, q) + \text{KL}(q(Z)\|p(Z|X,\Theta))
\end{aligned}
\tag{15}
$$

STEP 1: fix $\Theta = \Theta^{(g)}$, maximize $q(Z)$ for ELBO

$$
\begin{aligned}
q^*(\cdot) &= \arg\max_q \{\text{ELBO}(\Theta^{(g)}, q)\} \\
&= \arg\max_q \Big\{ \int_Z \log\left(\frac{p(X,Z|\Theta^{(g)})}{q(Z)}\right) q(Z)\mathrm{d}Z \Big\} \\
&= p(Z|X,\Theta^{(g)}) \\
\implies \text{ELBO}(\Theta^{(g)}, q^*) &= \mathcal{L}(\Theta^{(g)}|X)
\end{aligned}
\tag{16}
$$

This follows from the realization that when fixing $\Theta = \Theta^{(g)}$, the ELBO is upper-bounded by $\mathcal{L}(\Theta^{(g)}|X)$, and the maximum is attained exactly when $\text{KL}(q^* \| p(Z|X, \Theta^{(g)})) = 0$, i.e., $q^* = p(Z|X, \Theta^{(g)})$.

STEP 1 is invisible to the algorithm, but needs to be considered when interpreting convergence.

STEP 2: fix $q(\cdot) = p(Z|X, \Theta^{(g)})$, maximize Θ

substitute $q^* = p(Z|X, \Theta^{(g)})$ back into the ELBO:

$$
\begin{aligned}
\mathcal{L}(\Theta^{(g)}|X) &= \text{ELBO}(\Theta^{(g)}, q^*) \\
&= \int_Z \log\left(\frac{p(X, Z|\Theta^{(g)})}{p(Z|X, \Theta^{(g)})}\right) p(Z|X, \Theta^{(g)}) \mathrm{d}Z \\
&= \underbrace{\int_Z \log(p(X, Z|\Theta^{(g)})) p(Z|X, \Theta^{(g)}) \mathrm{d}Z}_{Q(\Theta^{(g)}|\Theta^{(g)})} \quad \underbrace{- \int_Z \log(p(Z|X, \Theta^{(g)})) p(Z|X, \Theta^{(g)}) \mathrm{d}Z}_{H(\Theta^{(g)}|\Theta^{(g)})}
\end{aligned} \tag{17}
$$

It is obvious that, after computing the E-M update from the decomposition in Eq.(17), we have:

$$
\begin{aligned}
\Theta^{(g+1)} &= \arg\max_{\Theta} Q(\Theta|\Theta^{(g)}) \\
&= \arg\max_{\Theta} \left(\int_Z \log\left(p(X, Z|\Theta)\right) p(Z|X, \Theta^{(g)}) \mathrm{d}Z \right)
\end{aligned} \tag{18}
$$

then:

$$
\underbrace{\int_Z \log(p(X, Z|\Theta^{(g+1)})) p(Z|X, \Theta^{(g)}) \mathrm{d}Z}_{Q(\Theta^{(g+1)}|\Theta^{(g)})} \geq \underbrace{\int_Z \log(p(X, Z|\Theta^{(g)})) p(Z|X, \Theta^{(g)}) \mathrm{d}Z}_{Q(\Theta^{(g)}|\Theta^{(g)})} \tag{19}
$$

but if we can also prove:

$$
\begin{aligned}
& H(\Theta|\Theta^{(g)}) \geq H(\Theta^{(g)}|\Theta^{(g)}) \qquad \forall \Theta \\
\implies \; & H(\Theta^{(g+1)}|\Theta^{(g)}) \geq H(\Theta^{(g)}|\Theta^{(g)})
\end{aligned} \tag{20}
$$

then we can show:

$$
\begin{aligned}
\mathcal{L}(\Theta^{(g+1)}|X) &= Q(\Theta^{(g+1)}|\Theta^{(g)}) + H(\Theta^{(g+1)}|\Theta^{(g)}) \\
&\geq \underbrace{Q(\Theta^{(g)}|\Theta^{(g)})}_{\text{Eq.(18)}} + \underbrace{H(\Theta^{(g)}|\Theta^{(g)})}_{\text{Eq.(20)}} \\
&= \mathcal{L}(\Theta^{(g)}|X)
\end{aligned} \tag{21}
$$

Note that since $Q(\Theta|\Theta^{(g)}) + H(\Theta|\Theta^{(g)}) = \mathcal{L}(\Theta|X)$ identically, jointly maximizing both terms would recover the (intractable) MLE itself. E-M maximizes only Q, so in general:

$$
\begin{aligned}
& \bar{\Theta} = \arg\max_{\Theta} \left\{ Q(\Theta|\Theta^{(g)}) + H(\Theta|\Theta^{(g)}) \right\} \\
\implies \; & \mathcal{L}(\Theta^{(g+1)}) \leq \mathcal{L}(\bar{\Theta})
\end{aligned} \tag{22}
$$

4.1.1 why $H(\Theta|\Theta^{(g)}) \geq H(\Theta^{(g)}|\Theta^{(g)}) \qquad \forall\Theta$

1. cross entropy

$$
\begin{aligned}
H(\Theta|\Theta^{(g)}) &= -\int_Z \log(p(Z|X,\Theta))p(Z|X,\Theta^{(g)})\mathrm{d}Z \\
&= \text{Cross-Entropy}\big(p(Z|X,\Theta^{(g)})\|p(Z|X,\Theta)\big) \\
\implies \arg\min_{\Theta}\{H(\Theta|\Theta^{(g)})\} &= \Theta^{(g)} \\
\implies \min_{\Theta}\{H(\Theta|\Theta^{(g)})\} &= \text{Entropy}(p(Z|X,\Theta^{(g)}))
\end{aligned}
\tag{23}
$$

2. directly

$$
\begin{aligned}
&H(\Theta|\Theta^{(g)}) - H(\Theta^{(g)}|\Theta^{(g)}) \\
&= \int_Z -\log(p(Z|X,\Theta))p(Z|X,\Theta^{(g)})\mathrm{d}Z - \int_Z -\log\big(p(Z|X,\Theta^{(g)})\big)p(Z|X,\Theta^{(g)})\mathrm{d}Z \\
&= \int_Z \log\left(\frac{p(Z|X,\Theta^{(g)})}{p(Z|X,\Theta)}\right)p(Z|X,\Theta^{(g)})\mathrm{d}Z \\
&= \int_Z -\log\left(\frac{p(Z|X,\Theta)}{p(Z|X,\Theta^{(g)})}\right)p(Z|X,\Theta^{(g)})\mathrm{d}Z \\
&\geq -\log\left(\int_Z \frac{p(Z|X,\Theta)}{p(Z|X,\Theta^{(g)})}p(Z|X,\Theta^{(g)})\mathrm{d}Z\right) \\
&= 0 \quad \because \phi = -\log \text{ is a convex function}
\end{aligned}
\tag{24}
$$

4.2 Proof of convergence without ELBO

let's decompose $\mathcal{L}(\Theta|X)$ into:

$$
\begin{aligned}
\mathcal{L}(\Theta|X) &= \log(p(X|\Theta)) \\
&= \log(p(Z,X|\Theta)) - \log(p(Z|X,\Theta))
\end{aligned}
\tag{25}
$$

take the expectation of both sides with respect to $p(Z|X,\Theta^{(g)})$:

$$
\begin{aligned}
\mathcal{L}(\Theta|X) &= \underbrace{\int_Z \log(p(Z,X|\Theta))p(Z|X,\Theta^{(g)})\mathrm{d}Z}_{Q(\Theta|\Theta^{(g)})} \underbrace{-\int_Z \log(p(Z|X,\Theta))p(Z|X,\Theta^{(g)})\mathrm{d}Z}_{H(\Theta|\Theta^{(g)})} \\
\implies \mathcal{L}(\Theta^{(g)}|X) &= \underbrace{\int_Z \log(p(Z,X|\Theta^{(g)}))p(Z|X,\Theta^{(g)})\mathrm{d}Z}_{Q(\Theta^{(g)}|\Theta^{(g)})} \underbrace{-\int_Z \log(p(Z|X,\Theta^{(g)}))p(Z|X,\Theta^{(g)})\mathrm{d}Z}_{H(\Theta^{(g)}|\Theta^{(g)})}
\end{aligned}
\tag{26}
$$

where $H(\Theta|\Theta^{(g)}) = \text{Cross-Entropy}\big(p(Z|X,\Theta^{(g)})\|p(Z|X,\Theta)\big)$.

Eq.(26) can be considered as a simpler version of the ELBO-KL decomposition of $\mathcal{L}(\Theta|X)$ we have seen previously in Eq.(17).

5 E-M Example: Gaussian Mixture Model

Gaussian Mixture Model (k-mixture) (GMM):

$$
\begin{aligned}
p(x|\Theta) = \sum_{l=1}^{k} \alpha_l \mathcal{N}(x|\mu_l, \Sigma_l) \qquad \sum_{l=1}^{k} \alpha_l = 1 \\
\text{and } \Theta = \{\alpha_1, \dots \alpha_k, \mu_1, \dots \mu_k, \Sigma_1, \dots \Sigma_k\}
\end{aligned}
\tag{27}
$$

For data $X = \{x_1, \dots x_n\}$ we introduce "latent" variables $Z = \{z_1, \dots z_n\}$, where each z_i indicates which mixture component x_i belongs to.

Looking at the E-M algorithm:

$$
\Theta^{(g+1)} = \arg\max_{\Theta} \left[q(\Theta, \Theta^{(g)}) \right] = \arg\max_{\Theta} \left(\int_Z \log\left(p(X, Z|\Theta)\right) p(Z|X, \Theta^{(g)}) \mathrm{d}Z \right) \tag{28}
$$

All we need to do is to define both $p(X, Z|\Theta)$ and $p(Z|X, \Theta)$

5.1 Gaussian Mixture Model in action

$$
p(X|\Theta) = \prod_{i=1}^{n} p(x_i|\Theta) = \prod_{i=1}^{n} \sum_{l=1}^{k} \alpha_l \mathcal{N}(x_i|\mu_l, \Sigma_l) \tag{29}
$$

How to define $p(X, Z|\Theta)$

$$
p(X, Z|\Theta) = \prod_{i=1}^{n} p(x_i, z_i|\Theta) = \prod_{i=1}^{n} \underbrace{p(x_i|z_i, \Theta)}_{\mathcal{N}(x_i|\mu_{z_i}, \Sigma_{z_i})} \underbrace{p(z_i|\Theta)}_{\alpha_{z_i}} = \prod_{i=1}^{n} \alpha_{z_i} \mathcal{N}(x_i|\mu_{z_i}, \Sigma_{z_i}) \tag{30}
$$

Notice that $p(X, Z|\Theta)$ is actually simpler than $p(X|\Theta)$.

How to define $p(Z|X, \Theta)$

$$
p(Z|X, \Theta) = \prod_{i=1}^{n} p(z_i|x_i, \Theta) = \prod_{i=1}^{n} \frac{\alpha_{z_i} \mathcal{N}(x_i|\mu_{z_i}, \Sigma_{z_i})}{\sum_{l=1}^{k} \alpha_l \mathcal{N}(x_i|\mu_l, \Sigma_l)} \tag{31}
$$

5.2 The E-Step

$$
\begin{aligned}
q(\Theta, \Theta^{(g)}) &= \int_Z \log\left(p(X, Z|\Theta)\right) p(Z|X, \Theta^{(g)}) \mathrm{d}Z \\
&= \int_{z_1} \cdots \int_{z_n} \left(\sum_{i=1}^{n} \log p(z_i, x_i|\Theta) \prod_{i=1}^{n} p(z_i|x_i, \Theta^{(g)}) \right) \mathrm{d}z_1 \cdots \mathrm{d}z_n
\end{aligned}
\tag{32}
$$

5.3 Some derivation to help

let $p(Y)$ be the joint pdf $p(y_1, \dots y_N)$, and let $F(Y)$ be an additively separable function, where each term involves only one variable y_i, i.e.,

$$
F(Y) = f_1(y_1) + \cdots + f_N(y_N) = \sum_{i=1}^{N} f_i(y_i) \tag{33}
$$

then,

$$
\int_{y_1} \cdots \int_{y_N} \Big(\sum_{i=1}^{N} f_i(y_i) \Big) p(Y) \mathrm{d}Y = \sum_{i=1}^{N} \Big(\int_{y_i} f_i(y_i) p(y_i) \mathrm{d}y_i \Big) \tag{34}
$$

5.3.1 Proof

$$\int_Y F(Y)p(Y)\mathrm{d}Y = \int_{y_1}\int_{y_2}\cdots\int_{y_N}\Big(\sum_{i=1}^N f_i(y_i)\Big)p(Y)\mathrm{d}y_1\cdots\mathrm{d}y_N \tag{35}$$

Expanding it out, this equation has N sum terms:

$$\begin{aligned}
&= \int_{y_1}\int_{y_2}\cdots\int_{y_N} f_1(y_1)p(y_1,\dots,y_N)\prod_{i=1}^N(\mathrm{d}y_i) + \cdots + \int_{y_1}\int_{y_2}\cdots\int_{y_N} f_N(y_N)p(y_1,\dots,y_N)\prod_{i=1}^N(\mathrm{d}y_i)\\
&= \int_{y_1} f_1(y_1)\mathrm{d}y_1\Big(\int_{y_2}\cdots\int_{y_N} p(y_1,\dots,y_N)\prod_{i=2}^N(\mathrm{d}y_i)\Big) + \cdots + \int_{y_N} f_N(y_N)\mathrm{d}y_N\int_{y_1}\cdots\int_{y_{N-1}} p(y_1,\dots,y_N)\prod_{i=1}^{N-1}(\mathrm{d}y_i)
\end{aligned} \tag{36}$$

the inner integral in the first big bracket is the marginal probability density $p(y_1)$; therefore, the first term becomes:

$$\int_{y_1} f_1(y_1)p(y_1)\mathrm{d}y_1 \tag{37}$$

Applying this to each of the N terms, therefore:

$$\int_Y F(Y)p(Y)\mathrm{d}Y = \int_{y_1} f_1(y_1)p(y_1)\mathrm{d}y_1 + \cdots + \int_{y_N} f_N(y_N)p(y_N)\mathrm{d}y_N \tag{38}$$

now apply Eq.(34), we have:

$$\begin{aligned}
q(\Theta,\Theta^{(g)}) &= \int_{z_1}\cdots\int_{z_n}\left(\sum_{i=1}^n \log p(z_i,x_i|\Theta)\prod_{i=1}^n p(z_i|x_i,\Theta^{(g)})\right)\mathrm{d}z_1\cdots\mathrm{d}z_n\\
&= \sum_{i=1}^n\left(\int_{z_i}\log p(z_i,x_i|\Theta)p(z_i|x_i,\Theta^{(g)})\mathrm{d}z_i\right) \qquad z_i\in\{1,\dots,k\}\\
&= \sum_{l=1}^k\sum_{i=1}^n \log p(z_i=l,x_i|\Theta)\,p(z_i=l|x_i,\Theta^{(g)}) \qquad \text{swap the summation order}\\
&= \sum_{l=1}^k\sum_{i=1}^n \log[\alpha_l\mathcal{N}(x_i|\mu_l,\Sigma_l)]p(l|x_i,\Theta^{(g)}) \qquad \text{substitute the Gaussian, write } z_i=l \text{ as } l
\end{aligned} \tag{39}$$

5.4 The M-Step objective function

$$\begin{aligned}
q(\Theta,\Theta^{(g)}) &= \sum_{l=1}^k\sum_{i=1}^n \log[\alpha_l\mathcal{N}(x_i|\mu_l,\Sigma_l)]p(l|x_i,\Theta^{(g)})\\
&= \sum_{l=1}^k\sum_{i=1}^n \log(\alpha_l)p(l|x_i,\Theta^{(g)}) + \sum_{l=1}^k\sum_{i=1}^n \log[\mathcal{N}(x_i|\mu_l,\Sigma_l)]p(l|x_i,\Theta^{(g)})
\end{aligned} \tag{40}$$

5.4.1 computing responsibilities

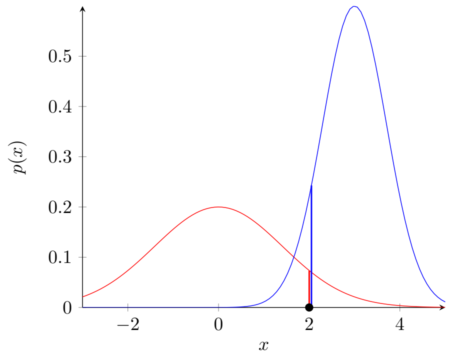


Figure 3: computing responsibility probabilities; red and blue each indicate their own unnormalized responsibilities

$$p(l|x_i, \Theta^{(g)}) = \frac{\alpha_l \mathcal{N}(\mathbf{x}_i|\mu_l, \Sigma_l)}{\sum_{s=1}^{k} \alpha_s \mathcal{N}(\mathbf{x}_i|\mu_s, \Sigma_s)} \tag{41}$$

Eq.(40) shows that the first term contains only α and the second term contains only μ, Σ. So we can maximize both terms independently.

5.5 The M-Step: maximizing α

Maximizing α means that:

$$\frac{\partial \sum_{l=1}^{k} \sum_{i=1}^{n} \log(\alpha_l) p(l|x_i, \Theta^{(g)})}{\partial \alpha_1, \dots, \partial \alpha_k} = [0 \dots 0] \qquad \text{subject to } \sum_{l=1}^{k} \alpha_l = 1 \tag{42}$$

This is to be solved using Lagrange Multiplier

$$\mathbb{LM}(\alpha_1, \dots \alpha_k, \lambda) = \sum_{l=1}^{k} \log(\alpha_l) \underbrace{\left(\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)}) \right)}_{\text{contains no } \alpha} + \lambda \left(\sum_{l=1}^{k} \alpha_l - 1 \right) \tag{43}$$

taking derivative with respect to just one α_l:

$$\begin{aligned}
\frac{\partial \mathbb{LM}}{\partial \alpha_l} &= \frac{1}{\alpha_l} \left(\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)}) \right) + \lambda = 0 \\
\implies -\lambda &= \frac{1}{\alpha_l} \left(\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)}) \right) \\
\implies -\lambda \alpha_l &= \left(\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)}) \right) \\
\implies -\lambda \sum_{l=1}^{k} \alpha_l &= \sum_{l=1}^{k} \left(\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)}) \right) \\
\implies \lambda &= -n
\end{aligned} \tag{44}$$

substitute $\lambda = -n$ into Eq.(44):

$$\begin{aligned}
&\frac{1}{\alpha_l}\left(\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)})\right) + \lambda = 0 \\
\Longrightarrow\ &\frac{1}{\alpha_l}\left(\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)})\right) - n = 0 \\
&\qquad\qquad\qquad \Longrightarrow\ \alpha_l = \frac{1}{n}\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)})
\end{aligned} \tag{45}$$

5.6 The M-Step: maximizing μ, Σ

Here I jot down the MLE steps for the parameters of a single multidimensional Gaussian distribution. The MLE of a single 1D Gaussian distribution can be easily found in your previous work.

So, maximizing μ, Σ means that:

$$\frac{\partial \sum_{l=1}^{k} \sum_{i=1}^{n} \log[\mathcal{N}(x_i|\mu_l, \Sigma_l)] p(l|x_i, \Theta^{(g)})}{\partial \mu_1, \dots, \partial \mu_k, \partial \Sigma_1, \dots, \partial \Sigma_k} = [0 \dots 0] \tag{46}$$

You will need some linear algebra identities to solve this; it is quite involved. For full details, refer to Bilmes' gentle tutorial [5].

5.6.1 Some formulas to remember

- derivatives of log of determinant (**with** determinant)

$$\frac{\partial \log |\mathbf{X}|}{\partial \mathbf{X}} = (\mathbf{X}^{-1})^\top \tag{47}$$

- Derivatives of Traces

$$\frac{\partial \text{Tr}(F(\mathbf{X}))}{\partial \mathbf{X}} = (f(\mathbf{X}))^\top \tag{48}$$

where $f(\cdot)$ is the scalar derivative of $F(\cdot)$

- Derivatives of Traces, fact 1

$$\frac{\partial \text{Tr}(\mathbf{A}\mathbf{X}\mathbf{B})}{\partial \mathbf{X}} = \mathbf{A}^\top \mathbf{B}^\top \tag{49}$$

- Derivatives of Traces of inverse, fact 2

$$\frac{\partial \text{Tr}((\mathbf{X}+\mathbf{A})^{-1})}{\partial \mathbf{X}} = -\big((\mathbf{X}+\mathbf{A})^{-1}(\mathbf{X}+\mathbf{A})^{-1}\big)^\top \tag{50}$$

- Derivatives of Traces of inverse, fact 3

$$\frac{\partial \text{Tr}(\mathbf{A}\mathbf{X}^{-1}\mathbf{B})}{\partial \mathbf{X}} = -(\mathbf{X}^{-1}\mathbf{B}\mathbf{A}\mathbf{X}^{-1})^\top \tag{51}$$

5.6.2 Maximization μ_l

$$
\begin{aligned}
\text{second part of } q(\Theta, \Theta^{(g)}) &= \sum_{l=1}^{k} \sum_{i=1}^{n} \log[\mathcal{N}(x_i | \mu_l, \mathbf{\Sigma}_l)] p(l | x_i, \Theta^{(g)}) \\
&= \sum_{i=1}^{n} \sum_{l=1}^{k} \log\left(\frac{1}{\sqrt{(2\pi)^d |\mathbf{\Sigma}_l|}} \exp\left(-\frac{1}{2}(x_i - \boldsymbol{\mu}_l)^\top \mathbf{\Sigma}_l^{-1} (x_i - \boldsymbol{\mu}_l) \right) \right) p(l | x_i, \Theta^{(g)})
\end{aligned} \tag{52}
$$

As a template, recall the log-likelihood of N i.i.d. zero-meaned columns $\mathbf{Y}$ (each column in $\mathbb{R}^D$) under $\mathcal{N}(\mathbf{0}, \mathcal{K})$:

$$
\mathcal{L} \equiv \mathcal{L}(p(\mathbf{Y}|\mathcal{K})) = -\frac{DN}{2}\log(2\pi) - \frac{N}{2}\log|\mathcal{K}| - \frac{1}{2}\mathrm{Tr}\big(\mathcal{K}^{-1}\mathbf{Y}\mathbf{Y}^\top\big) \tag{53}
$$

Let $\mathbf{Y}$ be the zero-meaned data matrix, where each column of $\mathbf{Y}$ is $\mathbf{x}_i - \boldsymbol{\mu}_l$:

$$
\mathcal{S}(\mu_l, \mathbf{\Sigma}_l) = \sum_{i=1}^{n} -\frac{1}{2}\log(|\mathbf{\Sigma}_l|) p(l|x_i, \Theta^{(g)}) - \sum_{i=1}^{n} \frac{1}{2}(x_i - \boldsymbol{\mu}_l)^\top \mathbf{\Sigma}_l^{-1} (x_i - \boldsymbol{\mu}_l) p(l|x_i, \Theta^{(g)}) + \text{Const.} \tag{54}
$$

$$
\begin{aligned}
\implies \mathcal{S}(\mu_l, \Sigma_l^{-1}) &= -\mathrm{Tr}\left(\frac{\Sigma_l^{-1}}{2} \sum_{i=1}^{n} (x_i - \mu_l)(x_i - \mu_l)^\top p(l|x_i, \Theta^{(g)}) \right) + \text{ Constant} \\
\implies \frac{\partial \mathcal{S}(\mu_l, \Sigma_l^{-1})}{\partial \mu_l} &= \frac{\Sigma_l^{-1}}{2} \sum_{i=1}^{n} 2(x_i - \mu_l) p(l|x_i, \Theta^{(g)}) = 0 \\
\implies \sum_{i=1}^{n} x_i p(l|x_i, \Theta^{(g)}) &= \mu_l \sum_{i=1}^{n} p(l|x_i, \Theta^{(g)}) \\
\implies \mu_l &= \frac{\sum_{i=1}^{n} x_i p(l|x_i, \Theta^{(g)})}{\sum_{i=1}^{n} p(l|x_i, \Theta^{(g)})}
\end{aligned} \tag{55}
$$

5.6.3 Maximization of covariance

$$
\begin{aligned}
\text{second part of } q(\Theta, \Theta^{(g)}) &= \sum_{l=1}^{k} \sum_{i=1}^{n} \log[\mathcal{N}(x_i | \mu_l, \mathbf{\Sigma}_l)] p(l | x_i, \Theta^{(g)}) \\
&= \sum_{i=1}^{n} \sum_{l=1}^{k} \log\left(\frac{1}{\sqrt{(2\pi)^d |\mathbf{\Sigma}_l|}} \exp\left(-\frac{1}{2}(x_i - \boldsymbol{\mu}_l)^\top \mathbf{\Sigma}_l^{-1} (x_i - \boldsymbol{\mu}_l) \right) \right) p(l | x_i, \Theta^{(g)})
\end{aligned} \tag{56}
$$

- let $\mathbf{Y}$ be zero-meaned data matrix, where each column of $\mathbf{Y}$ is $x_i - \boldsymbol{\mu}_l$
- let $\mathbf{P}$ be diagonal matrix in which $\mathbf{P}_{ii}$ correspond to $p(l|x_i, \Theta^{(g)})$

$$
\mathcal{L} \equiv \mathcal{L}(p(\mathbf{Y}|\boldsymbol{\mu}_l, \mathbf{\Sigma}_l)) = -\frac{d \times \mathrm{Tr}(\mathbf{P})}{2}\log(2\pi) - \frac{\mathrm{Tr}(\mathbf{P})}{2}\log|\mathbf{\Sigma}_l| - \frac{1}{2}\mathrm{Tr}\big(\mathbf{\Sigma}_l^{-1}\mathbf{Y}\mathbf{P}\mathbf{Y}^\top\big) \tag{57}
$$

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{\Sigma}_l} &= \mathbf{\Sigma}_l{}^{-1}\mathbf{Y}\mathbf{P}\mathbf{Y}^\top\mathbf{\Sigma}_l^{-1} - \text{Tr}(\mathbf{P})\mathbf{\Sigma}_l^{-1} = \mathbf{0} \\
\implies \mathbf{\Sigma}_l{}^{-1}\mathbf{Y}\mathbf{P}\mathbf{Y}^\top\mathbf{\Sigma}_l^{-1} &= \text{Tr}(\mathbf{P})\mathbf{\Sigma}_l^{-1} \\
\implies \mathbf{Y}\mathbf{P}\mathbf{Y}^\top\mathbf{\Sigma}_l^{-1} &= \text{Tr}(\mathbf{P})\,\mathbf{I} \\
\implies \mathbf{\Sigma}_l^{-1} &= \text{Tr}(\mathbf{P})(\mathbf{Y}\mathbf{P}\mathbf{Y}^\top)^{-1} \\
\implies \mathbf{\Sigma}_l &= \text{Tr}(\mathbf{P})^{-1}(\mathbf{Y}\mathbf{P}\mathbf{Y}^\top) = \frac{(\mathbf{Y}\mathbf{P}\mathbf{Y}^\top)}{\text{Tr}(\mathbf{P})} \\
&= \frac{\sum_{i=1}^n (x_i-\mu_l)(x_i-\mu_l)^\top p(l|x_i,\Theta^{(g)})}{\sum_{i=1}^n p(l|x_i,\Theta^{(g)})}
\end{aligned} \tag{58}$$

$$\mathcal{S}(\mu_l, \Sigma_l^{-1}) = \sum_{i=1}^n \left(-\frac{1}{2}\log(|\Sigma_l|) - \frac{1}{2}(x_i-\mu_l)^\top \Sigma_l^{-1}(x_i-\mu_l) \right) p(l|x_i,\Theta^{(g)}) \tag{59}$$

Change Σ to Σ^{-1}, this is so that after taking derivative of $\log(X)$, the result is in terms of X^{-1}

$$\begin{aligned}
&= \left(\frac{1}{2}\sum_{i=1}^n \log(|\Sigma_l^{-1}|) p(l|x_i,\Theta^{(g)}) - \frac{1}{2}\text{tr}\left(\Sigma_l^{-1} \underbrace{\sum_{i=1}^n (x_i-\mu_l)(x_i-\mu_l)^\top p(l|x_i,\Theta^{(g)})}_{M_l} \right) \right) \\
\implies \frac{\partial \mathcal{S}(\mu_l,\Sigma_l^{-1})}{\partial \Sigma_l^{-1}} &= \frac{2\sum_{i=1}^n \Sigma_l p(l|x_i,\Theta^{(g)}) - \sum_{i=1}^n \text{diag}(\Sigma_l) p(l|x_i,\Theta^{(g)})}{2} - \frac{2M_l - \text{diag}(M_l)}{2} = 0 \\
&\implies 2\Big(\sum_{i=1}^n \Sigma_l p(l|x_i,\Theta^{(g)}) - M_l \Big) - \text{diag}\Big(\sum_{i=1}^n \Sigma_l p(l|x_i,\Theta^{(g)}) - M_l \Big) = 0 \\
&\implies \sum_{i=1}^n \Sigma_l p(l|x_i,\Theta^{(g)}) - M_l = 0 \\
&\implies \Sigma_l = \frac{M_l}{\sum_{i=1}^n p(l|x_i,\Theta^{(g)})} = \frac{\sum_{i=1}^n (x_i-\mu_l)(x_i-\mu_l)^\top p(l|x_i,\Theta^{(g)})}{\sum_{i=1}^n p(l|x_i,\Theta^{(g)})}
\end{aligned} \tag{60}$$

5.7 Summary of Gaussian Mixture Model

In summary, to update $\Theta^{(g)} \to \Theta^{(g+1)}$:

$$\alpha_l^{(g+1)} = \frac{1}{n}\sum_{i=1}^n p(l|x_i,\Theta^{(g)}) \tag{61}$$

$$\mu_l^{(g+1)} = \frac{\sum_{i=1}^n x_i p(l|x_i,\Theta^{(g)})}{\sum_{i=1}^n p(l|x_i,\Theta^{(g)})} \tag{62}$$

$$\Sigma_l^{(g+1)} = \frac{\sum_{i=1}^n [x_i-\mu_l^{(g+1)}][x_i-\mu_l^{(g+1)}]^\top p(l|x_i,\Theta^{(g)})}{\sum_{i=1}^n p(l|x_i,\Theta^{(g)})} \tag{63}$$

One can verify that when the number of components $k=1 \implies p(l=1|x_i,\Theta^{(g)})=1$, then:

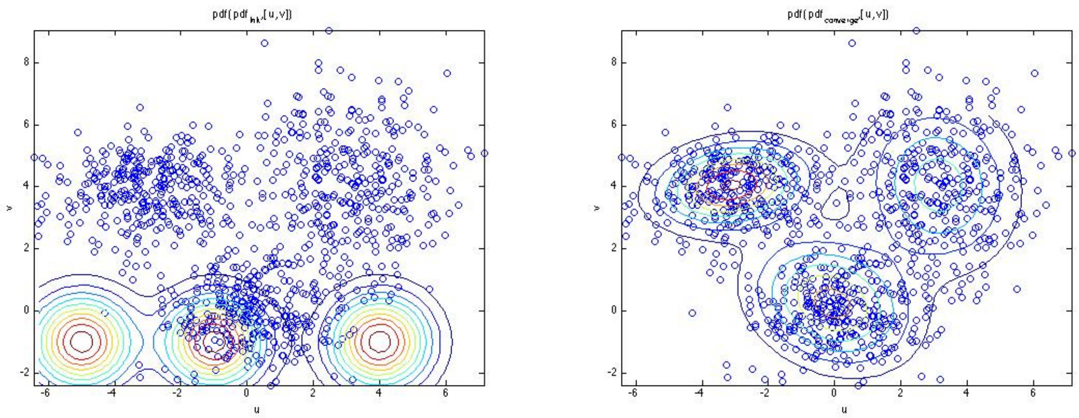
$$\alpha_l^{(g+1)} = 1 \tag{64}$$

$$\mu_l^{(g+1)} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{65}$$

$$\Sigma_l^{(g+1)} = \frac{1}{n}\sum_{i=1}^{n}[x_i - \mu_l^{(g+1)}][x_i - \mu_l^{(g+1)}]^\top \tag{66}$$

which becomes the parameter update of a single Gaussian

5.8 GMM fitting: initialization vs. convergence



(a) GMM at initialization: $\Theta^{(1)}$

(b) GMM at convergence: $\Theta^{(\text{final})}$

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